The above information may be represented in the form of a 3×5 matrix A as

$$A = \begin{bmatrix} 48 & 71 & 80 & 62 & 55 \\ 70 & 68 & 91 & 73 & 60 \\ 77 & \mathbf{84} & 95 & 82 & 62 \end{bmatrix}.$$

The entry 84 common to the third row and the second column in the matrix represents the mark scored by the student in English Exam 3.

Example 7.1

Suppose that a matrix has 12 elements. What are the possible orders it can have? What if it has 7 elements?

Solution

The number of elements is the product of number of rows and number of columns. Therefore, we will find all ordered pairs of natural numbers whose product is 12. Thus, all the possible orders of the matrix are $1\times12,\ 12\times1,\ 2\times6,\ 6\times2,\ 3\times4$ and 4×3.

Since 7 is prime, the only possible orders of the matrix are 1×7 and 7×1.

Example 7.2

Construct a 2×3 matrix whose $(i, j)^{\text{th}}$ element is given by

$$a_{ij} = \frac{\sqrt{3}}{2}|2i - 3j| \quad (1 \le i \le 2,\ 1 \le j \le 3).$$

Solution

In general, a 2×3 matrix is given by $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

By definition of a_{ij}, we easily have $a_{11} = \frac{\sqrt{3}}{2}|2-3| = \frac{\sqrt{3}}{2}$ and other entries of the matrix A may be computed similarly. Thus, the required matrix A is $\begin{bmatrix} \frac{\sqrt{3}}{2} & 2\sqrt{3} & \frac{7\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \sqrt{3} & \frac{5\sqrt{3}}{2} \end{bmatrix}$.

7.2.1 Types of Matrices

Row, Column, Zero matrices

Definition 7.2

A matrix having only one row is called a **row matrix**.

For instance, $A = [A]_{1\times4} = [1 \quad 0 \quad -1.1 \quad \sqrt{2}]$ is a row matrix. More generally, $A = [a_{ij}]_{1\times n} = [a_{1j}]_{1\times n}$ is a row matrix of order $1\times n$.

Definition 7.3

A matrix having only one column is called a **column matrix**.

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